# Skills Summary

## Choosing a Right-Triangle Method

*Use this reference while completing your worksheet or when studying.*

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**The decision rule.** Look at what is GIVEN, not at what is asked. Angle given or wanted? If no, Pythagoras. If yes and the angle is 30°, 45° or 60°, a special triangle. Otherwise a trig ratio.

## 1. Pythagorean theorem: two sides, no angle

**Rule:** in a right triangle with the right angle at $C$,

$$a^2 + b^2 = c^2 \qquad (c \text{ is always the hypotenuse})$$

**Use it when** two sides are known and the third is wanted, and no angle appears anywhere.
**Missing leg:** rearrange to $b^2 = c^2 - a^2$ — subtract, never add.

**Sanity check:** the hypotenuse must come out largest. Worth memorising: 3-4-5, 5-12-13, 8-15-17 and their multiples.

**Example:** The legs of a right triangle are 9 and 12. Find the hypotenuse.
**Step 1.** Two sides are given and no angle appears, so a trig ratio has nothing to work with — Pythagoras is the only tool that fits.
**Step 2.** Both given sides are legs, so add: $9^2 + 12^2 = 81 + 144 = 225$, and the hypotenuse is $\sqrt{225}$. $c =$ **15**

**Try it:** A right triangle has hypotenuse 20 and one leg 16. Find the other leg.

check: **12**

## 2. Special triangles: 45-45-90 and 30-60-90

**Two ratios worth knowing exactly:**

- 45°-45°-90°: leg : leg : hypotenuse $= 1 : 1 : \sqrt{2}$. Hypotenuse = leg $\times\sqrt{2}$.
- 30°-60°-90°: short leg : long leg : hypotenuse $= 1 : \sqrt{3} : 2$. The short leg is *opposite* the 30° angle and is half the hypotenuse.

**Use it when** an acute angle is 30°, 45° or 60° — you get an exact answer with no calculator. A square's diagonal and an equilateral triangle's altitude always create these.

**Example:** An isosceles right triangle has legs of 8. Find the hypotenuse.
**Step 1.** Isosceles and right forces the two acute angles to be equal, which is the special ratio case — exact, no calculator work needed.
**Step 2.** Hypotenuse = leg $\times\sqrt{2} = 8\sqrt{2}$, which is exact; as a decimal it is $8 \times 1.41$. $c = 8\sqrt{2} \approx$ **11.31**

**Try it:** A right triangle has hypotenuse 14 and an acute angle of 30°. Find the side opposite that angle.

check: **7**

## 3. Trig ratios: when an angle is in play

**SOH-CAH-TOA**, always taken *relative to the angle you are using*:

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \qquad \cos A = \frac{\text{adj}}{\text{hyp}}, \qquad \tan A = \frac{\text{opp}}{\text{adj}}$$

**Side missing:** pick the ratio containing the two sides you care about (one known, one wanted), then multiply or divide.

**Angle missing:** form the same ratio from the two known sides, then apply the inverse. Example: adjacent 6, hypotenuse 10 gives $\cos A = 0.6$, so $A = \cos^{-1}(0.6) \approx 53.13°$.

Choosing the ratio is the whole skill: the one you want is the one whose two sides are the two you have.

**Example:** A right triangle has hypotenuse 17 and an acute angle of 28°. Find the side opposite that angle.
**Step 1.** The angle is not one of the special ones, so the exact ratios are out; the wanted side is opposite the known angle and the known side is the hypotenuse, and that pairing is sine.
**Step 2.** $\sin 28° = \frac{\text{opp}}{17}$, so the opposite side is $17 \sin 28°$. opp = **7.98**

**Try it:** A right triangle has hypotenuse 24 and an acute angle of 41°. Find the side *adjacent* to that angle.

check: **18.11**

**(!) Watch out:** "opposite" and "adjacent" are named relative to the angle you are using, so they swap when you switch angles — the hypotenuse never does. And if the missing quantity is an angle, the answer comes out of an inverse ratio; forgetting the inverse returns a ratio between −1 and 1 instead of a number of degrees.